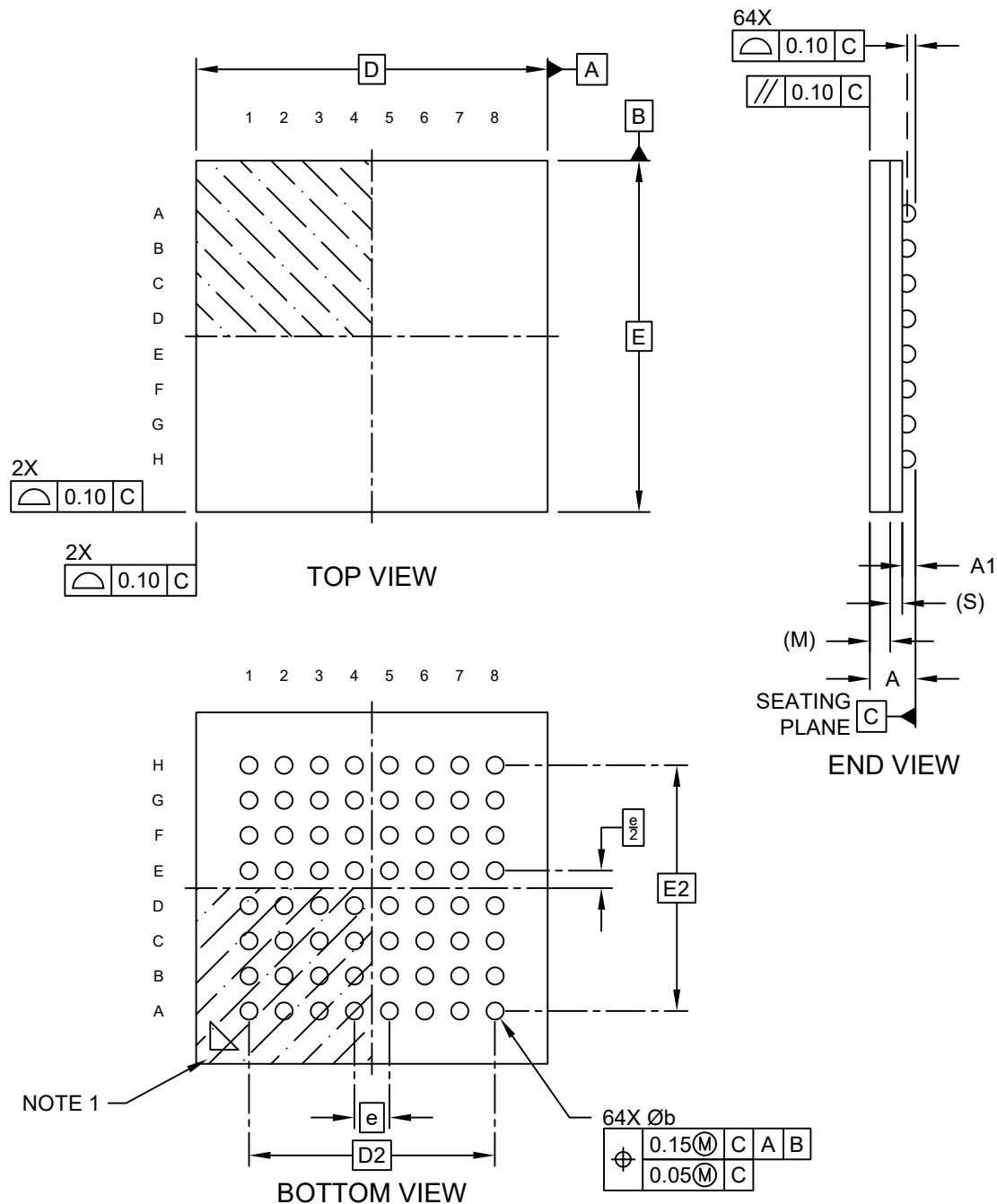


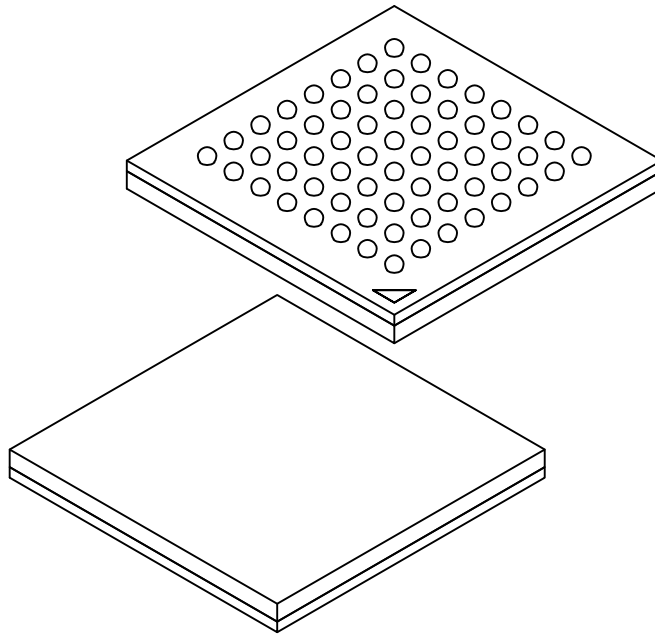
64-Ball Ultra Thin Fine-Pitch Ball Grid Array Package (BQB) - 5x5x0.65 mm Body [UFBGA]; Atmel Legacy Global Package Code CAH

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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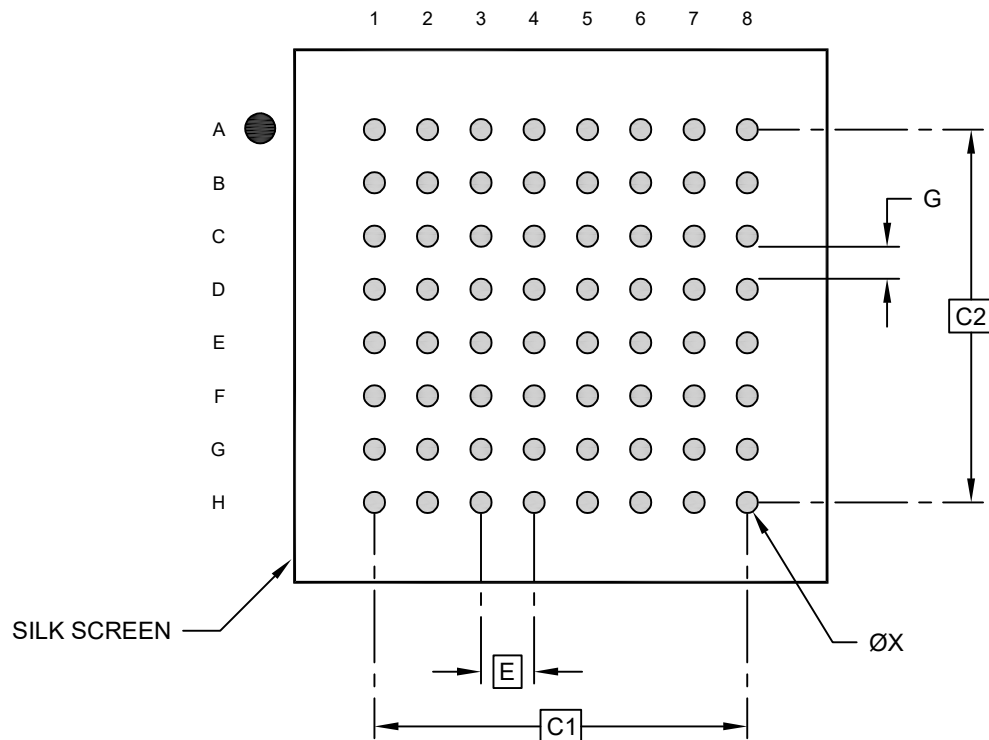
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	–	–	0.65
Ball Height	A1	0.14	0.19	0.24
Mold Thickness	M	0.25 REF		
Substrate Thickness	S	0.136 REF		
Overall Length	D	5.00 BSC		
Ball Array Length	D2	3.50 BSC		
Overall Width	E	5.00 BSC		
Ball Array Width	E2	3.50 BSC		
Ball Width	b	0.20	0.25	0.3

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

64-Ball Ultra Thin Fine-Pitch Ball Grid Array Package (BQB) - 5x5x0.65 mm Body [UFBGA]; Atmel Legacy Global Package Code CAH

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Contact Pad Spacing	C1		3.50 BSC	
Contact Pad Spacing	C2		3.50 BSC	
Contact Pad Width (Xnn)	X			0.20
Contact Pad to Contact Pad (Xnn)	G	0.30		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23153 Rev B